# 3. Analisis Fourier

- **Menganalisis suatu sinya**l adalah dengan meng-gunakan **deret Fourier.**
- Persamaan matematis suatu gelombang sinyal f(t),

$$f(t) = a_0 + \Sigma\, a_n \cos n\omega t + \Sigma\, b_n \operatorname{Sin} n\, \omega t]$$

Dimana : $a_0 = 1/T \int f(t)\, dt$

$a_n = 2/T \int f(t) \cos n\, \omega t\, dt$

$b_n = 2/T \int f(t) \sin n\, \omega t\, dt$

- $f(t) = a_0 + a_1 \operatorname{Cos} \omega t + a_2 \operatorname{Cos} 2\omega t + a_3 \operatorname{Cos} 3\omega t + \ldots\ldots\ldots$
  $+ b_1 \operatorname{Sin} \omega t + b_2 \operatorname{Sin} 2\omega t + b_3 \operatorname{Sin} 3\omega t + \ldots\ldots$

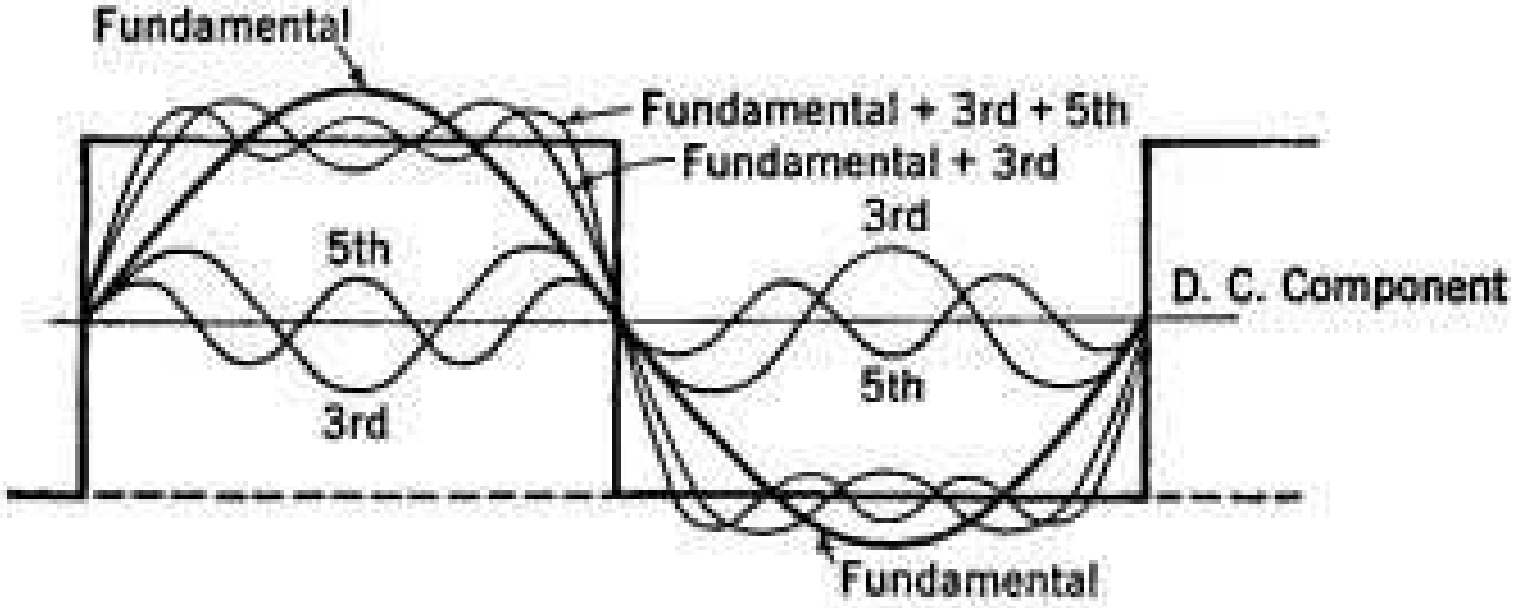
$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos n\omega t + \sum b_n \sin n\omega t]$$

Dimana : $a_0 = 1/T \int_0^T f(t)\ dt$

$$a_n = 2/T \int_0^T f(t) \cos n\omega t\ dt$$

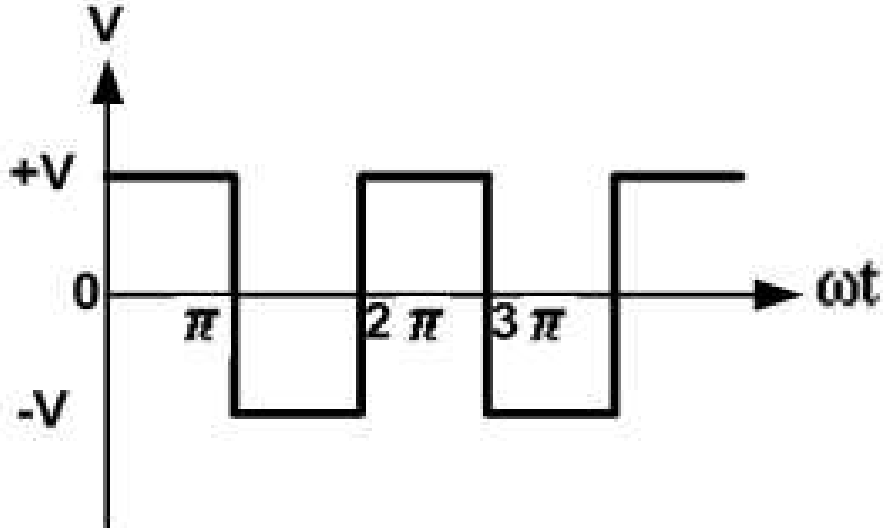
$$b_n = 2/T \int_0^T f(t) \sin n\omega t\ dt$$

- (f) = tegangan DC + sinyal sinusoida atau cosinsoida harmonisa orde pertama (fundamental) + sinyal sinusoida atau cosinsoida harmonisa orde kedua + sinyal sinusoida atau cosinsoida harmonisa orde ketiga + sinyal sinusoida atau cosinsoida harmonisa orde keempat +.....




**Penjumlahan Sinyal Sinusoida**

## Contoh : 1 Sinyal Bipolar




$f(t) = +V$ untuk $0 < \omega t < \pi$

$f(t) = -V$ untuk $\pi < \omega t < 2\pi$

$$a_0 = 1/T \int_0^T f(t)\ dt$$

$$= 0$$

$$a_n = 2/T \int_0^T f(t)\ \cos n\omega t\ d(\omega t)$$

$= 0$ ..... untuk setiap harga n

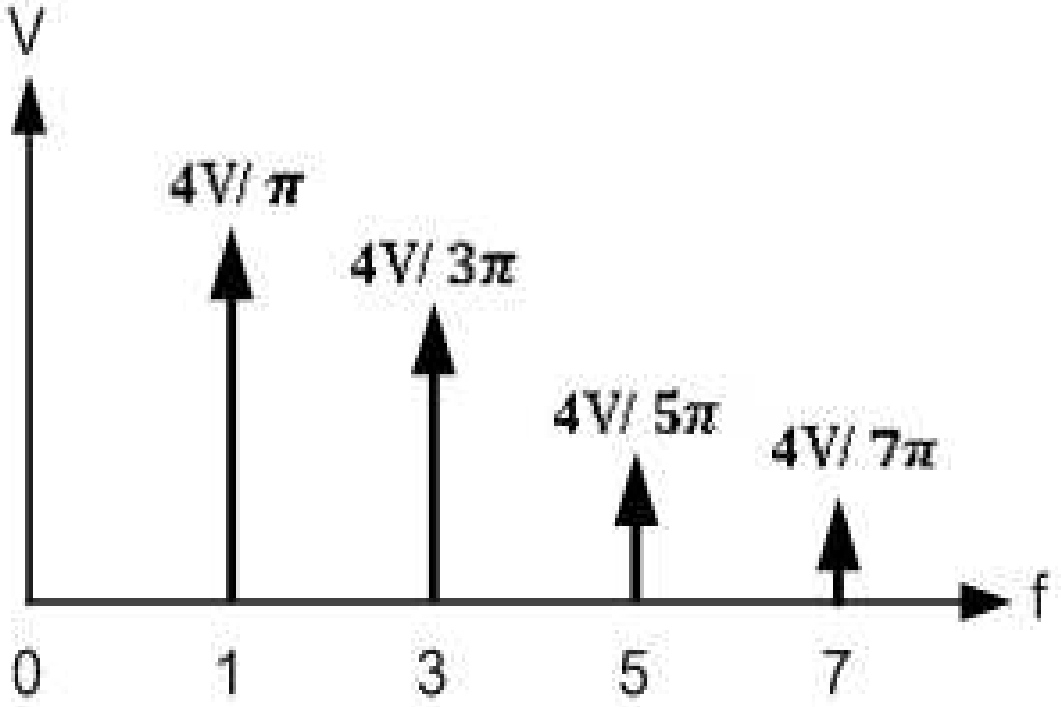
$$b_n = 2/T \int_0^T f(t)\ \text{Sin}\ n\omega t\ d(\omega t)$$

$= 4V/n\pi$.... untuk n ganjil

$= 0$ ..... untuk n genap

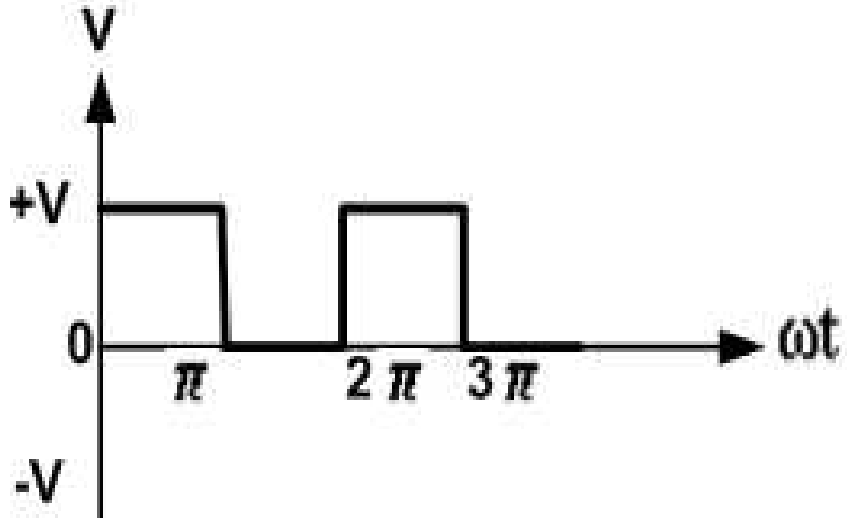
Jadi $f(t) = 4V/\pi \sin \omega t + 4V/3\pi \sin 3\omega t +$

$4V/5\pi \sin 5\omega t + 4V/7\pi \sin 7\omega t + .....$




**Gambar spektrumnya**

# Contoh : 2 Sinyal Unipolar




$f(t) = +V$ untuk $0 < \omega t < \pi$

$f(t) = 0$ untuk $\pi < \omega t < 2\pi$

$$a_0 = 1/T \int_0^T f(t)\ dt$$
$$= V/2$$

$$a_n = 2/T \int_0^T f(t)\ \cos n\omega t\ d(\omega t)$$
$$= 0 \quad \text{..... untuk setiap harga n}$$

$$b_n = 2/T \int_0^T f(t)\ \text{Sin}\ n\omega t\ d(\omega t)$$
$$= 2V/n\pi \text{.....} \quad \text{untuk n ganjil}$$
$$= 0 \text{.....} \quad \text{untuk n genap}$$

**f(t) = V/2 + 2V/$\pi$ Sin$\omega$t + 2V/3$\pi$ Sin3$\omega$t + 2V/5$\pi$ Sin5$\omega$t + .....**

